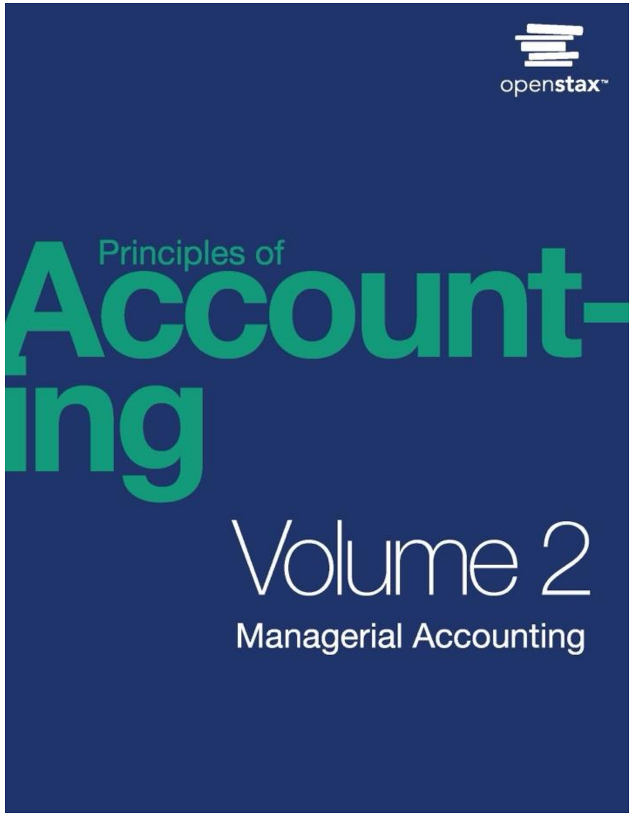
Principles of Accounting, Volume 2: Managerial Accounting

Chapter 8 STANDARD COSTS AND VARIANCES

PowerPoint Image Slideshow



Cost Accounting Subsystem

Because the cost of goods sold is likely to be the largest expense on a manufacturing firm's income statement, a critical part of the production process is an AIS's cost accounting subsystem. The **cost accounting subsystem** provides important control information (e.g., variance reports reflecting differences between actual and standard production costs) and varies with the size of the company and the types of product produced. As you might guess, a bakery producing baked goods would collect very different data in its AIS compared to an automobile manufacturer. Cost accounting subsystems for manufacturing organizations are commonly job costing, process costing, or activity-based costing systems.

Chapter Outline

- **8.1 Explain How and Why a Standard Cost Is Developed**
- 8.2 Compute and Evaluate Materials Variances
- 8.3 Compute and Evaluate Labor Variances
- 8.4 Compute and Evaluate Overhead Variances
- 8.5 Describe How Companies Use Variance Analysis

- Fundamentals of standard costs
  - Companies have an expectation of cost and quantity used, or a **standard**, for production or providing a service.
  - A **standard cost** is an expected cost established at the beginning of a fiscal year and is an expected amount paid for materials costs or labor rates. The standard quantity is the expected usage amount of materials or labor.
  - The company can then compare the standard costs against its actual results to measure its efficiency.

- Manufacturing cost variances
  - Once a company determines a standard cost, they then evaluate any variances or differences that occur when comparing actual results to standard (expected) results.
  - Variances are either favorable or unfavorable.

Figure 8.2

STANDARD COST CARD					
Product: 1 Cup of Coffee					
Manufacturing Cost Information	Standard Quantity	×	Standard Cost per Unit	=	Cost Summary
Direct Materials					
Material (coffee grounds)	ounces		per ounce		
Direct Labor					
Barista	minute		per minute		
Manufacturing Overhead					
Variable overhead	minute		per direct labor minute		
Fixed overhead	minute		per direct labor minute		
Standard Cost per cup of coffee					

Standard Cost Card for a Coffee Shop.

In-class Exercise

EA2. Rene is working with the operations manager to determine what the standard labor cost is for a spice chest. He has watched the process from start to finish and taken detailed notes on what each employee does. The first employee selects and mills the wood, so it is smooth on all four sides. This takes the employee 1 hour for each chest. The next employee takes the wood and cuts it to the proper size. This takes 30 minutes. The next employee assembles and sands the chest. Assembly takes 2 hours. The chest then goes to the finishing department. It takes 1.5 hours to finish the chest. All employees are cross-trained so they are all paid the same amount per hour, $17.50.

A. What are the standard hours per chest?

B. What is the standard cost per chest for labor?

Your Turn: Developing a Standard Cost Card

Use the information provided to create a standard cost card for production of one deluxe bicycle from Bicycles Unlimited.

To make one bicycle it takes four pounds of material. The material can usually be purchased for $5.25 per pound. The labor necessary to build a bicycle consists of two types. The first type of labor is assembly, which takes 2.75 hours. These workers are paid $11.00 per hour. The second type of labor is finishing, which takes 4 hours. These workers are paid $15.00 per hour. Overhead is applied using labor hours. The variable overhead rate is $5.00 per labor hour. The fixed overhead rate is $3.00 per hour.

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Total Direct Materials Variance

Direct Materials ________ Variance

Direct Materials _________ Variance

Total Direct Labor Variance

Direct Labor ______ Variance

Direct Labor ______ Variance

Example: Direct Materials Price Variance & Quantity Variance

Connie's Candy Company produces various types of candies that they sell to retailers. Connie's Candy establishes a standard price for candy-making materials of $7.00 per pound. Each box of candy is expected to use 0.25 pounds of candy-making materials. Connie's Candy found that the actual price of materials was $9.00 per pound and that the actual quantity of candy-making materials used to produce one box of candy was 0.50 per pound. Compute direct materials price variance, direct materials quantity variance, and total direct materials variance.

Total Direct Materials Variance

Direct Materials ________ Variance

Direct Materials ________ Variance

Example. Sweet and Fresh Shampoo Materials

Biglow Company makes a hair shampoo called Sweet and Fresh. Each bottle has a standard material cost of 8 ounces at $0.85 per ounce. During May, Biglow manufactured 11,000 bottles. They bought 89,000 ounces of material at a cost of $74,760. All 89,000 ounces were used to make the 11,000 bottles. Calculate the material price variance and the material quantity variance.

Total Direct Materials Variance

Direct Materials ________ Variance

Direct Materials ________ Variance

In-class Exercise

EA 6. Use the information provided to answer the questions.

Actual price paid per pound of material	$ 14.50
Total standard pounds for units produced this period	12,500
Pounds of material used	13,250
Direct materials price variance favorable	$4,637.50

All material purchased was used in production.

A. What is the standard price paid for materials?
B. What is the direct materials quantity variance?
C. What is the total direct materials cost variance?
D. If the direct materials price variance was unfavorable, what would be the standard price?

In-class Exercise

EA 6. Use the information provided to answer the questions.

Actual price paid per pound of material	$ 14.50
Total standard pounds for units produced this period	12,500
Pounds of material used	13,250
Direct materials price variance favorable	$4,637.50

All material purchased was used in production

PA 4. April Industries employs a standard costing system in the manufacturing of its sole product, a park bench. They purchased 60,000 feet of raw material for $300,000, and it takes 5 feet of raw materials to produce one park bench. In August, the company produced 10,000 park benches. The standard cost for material output was $100,000, and there was an unfavorable direct materials quantity variance of $6,000.

A. What is April Industries' standard price for one unit of material?
B. What was the total number of units of material used to produce the August output?
C. What was the direct materials price variance for August?

In-class Exercise

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A. What is April Industries’ standard price for one unit of material?

PA 4. April Industries employs a standard costing system in the manufacturing of its sole product, a park bench. They purchased 60,000 feet of raw material for $300,000, and it takes 5 feet of raw materials to produce one park bench. In August, the company produced 10,000 park benches. The standard cost for material output was $100,000, and there was an unfavorable direct materials quantity variance of $6,000.

B. total # of units of material used to produce the August output?

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C. What was the direct materials price variance for August?

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Example: Direct Labor Rate Variance

Connie’s Candy establishes a standard rate per hour for labor of $8.00. Each box of candy is expected to require 0.10 hours of labor (6 minutes). Connie’s Candy found that the actual rate of pay per hour for labor was $9.50. Actual labor to make each box was 0.20 hours.

A. Calculate the direct labor rate variance. Is this variance favorable or unfavorable?

B. Calculate the direct labor time variance. Is this variance favorable or unfavorable?

Direct Labor Variance Tree

Total Direct Labor Variance

Direct Labor ______ Variance

Direct Labor ______ Variance

In-class Exercise

Barley, Inc., produces a product and has the following as standard costs per unit for materials and labor:

Materials	4 pounds @ $15 per pound
Labor	2 hours @ $20 per hour

For the month of October, the following information was gathered related to production:

Beginning inventory	0
Units completed	10,000
Budgeted output units	12,000
Materials used (50,000 pounds)	$800,000
Labor (25,000 hours)	$450,000

Compute:

A. The materials price and quantity variances

B. The labor rate and efficiency variances

PA 8. Breakaway Company’s labor information for May is as follows:

Actual direct labor hours worked	48,000
Standard direct labor hours allowed	47,400
Total payroll for direct labor	$1,128,000
Direct labor time variance	$ 13,800 (unfavorable)

A. What is the actual direct labor rate per hour?
B. What is the standard direct labor rate per hour?
C. What was the total standard direct labor cost for May?
D. What was the direct labor rate variance for May?

PA 8. Breakaway Company's labor information for May is as follows:

Actual direct labor hours worked	48,000
Standard direct labor hours allowed	47,400
Total payroll for direct labor	$1,128,000
Direct labor time variance	$ 13,800 (unfavorable)

A. What is the actual direct labor rate per hour?
B. What is the standard direct labor rate per hour?
C. What was the total standard direct labor cost for May?
D. What was the direct labor rate variance for May?

In-class Exercise

EA 12. Acme Inc. has the following information available:

Actual price paid for material	$1.00
Standard price for material	$1.20
Actual quantity purchased and used in production	100
Standard quantity for units produced	110
Actual labor rate per hour	$ 15
Standard labor rate per hour	$ 16
Actual hours	200
Standard hours for units produced	220

Compute the material price and quantity, and the labor rate and efficiency variances.

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Determination and evaluation of overhead variance

- Companies must compute a standard overhead rate in order determine their standard cost.

Total Variable (Manufacturing) Overhead (VOH) Cost Variance

VOH ______ Variance

VOH ___________ Variance

To determine the overhead standard cost, companies prepare a **flexible budget.**

- It gives estimated revenues and costs at varying levels of production.
- Note that at different levels of production, total fixed costs are the same, so the standard fixed cost per unit will change for each production level.
- Variable standard cost per unit is the same per unit for each level of production, but the total variable costs will change.

Static Quarterly Budget (Fig 7.24)

BIG BAD BIKES Static Quarterly Budget For Each Quarter	
Units Sold	1,500
Sales Price	$ 70
Sales	$ 105,000
Cost of Goods Sold	
Direct Material	6,000
Direct Labor	22,500
Variable Manufacturing Overhead	4,500
Fixed Manufacturing Overhead	29,000
Total Cost of Goods Sold	62,000
Gross Profit	43,000
Variable Sales and Admin	3,750
Fixed Sales and Admin	18,000
Interest Expense	0
Income Taxes	1,000
Total Other Expenses	$ 22,750
Net Income (Loss)	$ 20,250

Flexible Quarterly Budget (Fig. 7.25)

BIG BAD BIKES Flexible Budget For Year Ended December 31, 2019				
Units Sold		1,000	1,500	1,500
Sales Price		$ 70	$ 70	$ 75
Sales		$70,000	$105,000	$112,500
	Per-unit cost			
Cost of Goods Sold				
Direct Material	$ 4	$ 4,000	$ 6,000	$ 6,000
Direct Labor	15	15,000	22,500	22,500
Variable Manufacturing Overhead	3	3,000	4,500	4,500
Fixed Manufacturing Overhead		29,000	29,000	29,000
Total Cost of Goods Sold		51,000	62,000	62,000
Gross Profit		19,000	43,000	50,500
Variable Sales and Admin	2.50	2,500	3,750	3,750
Fixed Sales and Admin		18,000	18,000	18,000
Income Taxes		1,000	1,000	1,000
Total Other Expenses		21,500	22,750	22,750
Net Income (Loss)		$ (2,500)	$ 20,250	$ 27,750

These two columns are the same.

Static Budget for Big Bad Bikes. Modified for PPT.

Varying Production Levels for Big Bad Bikes. Modified for PPT.

Connie's Candy budgets capacity of production at 100% and determines expected overhead at this capacity.

Percent of capacity	90%	100%	110%
Direct labor hours	1,800	2,000	2,200
Units of output	900	1,000	1,100
Variable overhead	$3,600	$ 4,000	$ 4,400
Fixed overhead	$6,000	$ 6,000	$ 6,000
Total overhead	$9,600	$10,000	$10,400

Normal capacity = 100% and overhead is applied based on direct labor hours

Standard Overhead Rate = $10,000/2,000 = $5 per direct labor hour

Example

Flexible Budget Data

Percent of capacity	**100%**
Direct labor hours	2,000
Units of output	1,000
Variable overhead	$ 4,000
Fixed overhead	$ 6,000
Total overhead	$10,000

Actual Data

Percent of capacity	**100%**
Direct labor hours	2,500
Units of output	1,000
Variable overhead	$ 7,000
Fixed overhead	$ 6,000
Total overhead	$13,000

Effects of a Change in Actual Data

Flexible Budget Data

Percent of capacity	**100%**
Direct labor hours	2,000
Units of output	1,000
Variable overhead	$ 4,000
Fixed overhead	$ 6,000
Total overhead	$10,000

Actual Data

Percent of capacity	**100%**
Direct labor hours	1,800
Units of output	1,000
Variable overhead	$3,500
Fixed overhead	$6,000
Total overhead	$9,500

Total Variable Overhead (VOH) Cost Variance

VOH ______ Variance

VOH _________ Variance

In-class Exercise

EA11. A manufacturer planned to use $78 of variable overhead per unit produced, but in the most recent period, it actually used $76 of variable overhead per unit produced. During this same period, the company planned to produce 500 units but actually produced 540 units. What is the variable overhead spending variance?

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Module 8.5 Describe How Companies Use Variance Analysis

- When companies compare standards against actual results, they generally produce a variance report.
  - Managers often have to explain variances that are outside of a certain range.
  - Requiring managers to determine what caused unfavorable variances forces them to identify potential problem areas or consider if the variance was a one-time occurrence.
  - Requiring managers to explain favorable variances allows them to assess whether the favorable variance is sustainable.
  - Knowing what caused a variance allows management to plan for it in the future, depending on whether it was a one-time variance or will be ongoing.
  - Management can use standard costs to prepare the budget for the upcoming period.
  - Standard costs are a measurement tool and can therefore be used to evaluate performance.

REVIEW (1)

Standard Costs and Variances

- Standards are budgeted unit amounts for _______ paid and __________ used.
- Variances are the difference between __________ and __________ amounts.
- Variance can be favorable (________________) or unfavorable (_______________).
- All firms - manufacturing, retail, and service - use standards and variances

Direct Materials (DM) Variances

- Two components: DM ________ variance and DM __________ variance.
- The DM price variance is caused by _______ too much or too little for material.
- The DM quantity variance is caused by ________ too much or too little material.

REVIEW (2)

Direct Labor Variances

- Two components: direct labor _____ variance and direct labor _____ variance.
  - ➢ Actual labor rate vs. standard or budgeted labor rate
  - ➢ Actual labor hours vs. standard or budgeted labor hours

Variable Manufacturing Overhead (VMOH) Variances

- There are two sets of overhead variances: _________ and ______.
- VMOH: is it due to application rate differences or activity level differences?
  - ➢ Rate variance is the difference between the actual VMOH (=) and the VMOH that was expected given the number of hours worked (=).
  - ➢ Efficiency variance [=] is driven by the difference between the ______ hours worked and the _______ hours expected for the units produced.